



USN

## NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

### Fourth Semester B.E. (Biotechnology) (Credit System) Degree Examinations

Supplementary Examinations – December 2025

#### 21BT403 / 20BT403/17BT505 – ANALYTICAL TECHNIQUES

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

|    |                                                                                                                                                           | Unit – I   |     |     |     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|-----|-----|
|    |                                                                                                                                                           | Marks      | BT* | CO* | PO* |
| 1. | a) Instrumental analysis of a given sample involves several types of errors. What possible types of determinate errors occur in the analysis? Explain.    | 5          | L*1 | 1   | 1   |
|    | b) Illustrate calomel electrode.                                                                                                                          | 5          | L1  | 1   | 1   |
|    | c) Describe the method of external calibration.                                                                                                           | 10         | L2  | 2   | 2   |
| 2. | a) What are the different ways in which solution concentration is expressed?                                                                              | 5          | L1  | 1   | 1   |
|    | b) Obtain Henderson-Hasselbalch equation.                                                                                                                 | 5          | L3  | 1   | 2   |
|    | c) What are the different criteria used in the selection of best analytical method/instrumentation? Discuss.                                              | 10         | L2  | 2   | 2   |
| 3. | a) Calculate the ionic strength of:                                                                                                                       |            |     |     |     |
|    | i) 0.05 M NaOH                                                                                                                                            | 5          | L1  | 1   | 2   |
|    | ii) 1 M KCl                                                                                                                                               | 5          | L1  | 1   | 1   |
|    | b) What is the principle behind pH electrode?                                                                                                             |            |     |     |     |
|    | c) Explain about the significance of statistical parameters such as mean and standard deviation in analytical measurements by using any suitable example. | 10         | L2  | 2   | 2   |
|    |                                                                                                                                                           | Unit – II  |     |     |     |
| 4. | a) What is the principle behind UV-Vis spectroscopy? Describe its instrumentation.                                                                        | 10         | L2  | 3   | 2   |
|    | b) Depict the <sup>1</sup> H NMR spectrum of ethanol and describe the reasons for specific peak positions and the hyperfine splits.                       | 10         | L2  | 4   | 2   |
| 5. | a) Describe atomic absorption spectroscopy.                                                                                                               | 10         | L2  | 3   | 2   |
|    | b) How does MALDI-TOF help in protein identification? Explain.                                                                                            | 10         | L2  | 4   | 2   |
| 6. | a) Add a brief note on chromophores.                                                                                                                      | 5          | L2  | 3   | 1   |
|    | b) List a few applications of UV-Vis spectroscopy.                                                                                                        | 5          | L1  | 3   | 1   |
|    | c) Derive Larmour equation that governs NMR.                                                                                                              | 10         | L3  | 4   | 2   |
|    |                                                                                                                                                           | Unit – III |     |     |     |
| 7. | a) What is HPLC? How does it work? Explain.                                                                                                               | 10         | L2  | 5   | 2   |
|    | b) Can ion exchange chromatography be used for the separation of proteins? Discuss.                                                                       | 10         | L2  | 5   | 2   |
| 8. | a) How is DNA separation and analysis carried out using electrophoresis? Explain.                                                                         | 10         | L2  | 5   | 2   |
|    | b) Explain in detail about SDS-PAGE technique.                                                                                                            | 10         | L3  | 5   | 2   |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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